Task 1:
I01

```
(.venv) PS C:\Users\Do Duc Kiet\Downloads\BigData_Lab02_Group4> Get-Content .\task1\evidence\I01.json
{
  "test_id": "I01",
  "status": "PASS",
  "timestamp_utc": "2026-10-11T17:26:45.032156+00:00",
  "fixture_version": "research-telemetry-v1",
  "as_of": "2026-02-09T00:00:00Z",
  "bucket": "research-raw",
  "prefix": "lab2/inputs/batch-01/",
  "objects": [
    {
      "source_bucket": "research-raw",
      "source_key": "lab2/inputs/batch-01/observations_a.csv",
      "expected_bytes": 421028,
      "expected_sha256": "5157418915d867980fac8d84551a8b856bba1669ca2e33cef91a15ee82101aea",
      "status": "PASS",
      "http": 200,
      "actual_bytes": 421028,
      "actual_sha256": "5157418915d867980fac8d84551a8b856bba1669ca2e33cef91a15ee82101aea",
      "snapshot_path": "/tmp/lab2-i01/snapshot/observations_a.csv"
    },
    {
      "source_bucket": "research-raw",
      "source_key": "lab2/inputs/batch-01/observations_b.jsonl",
      "expected_bytes": 1037439,
      "expected_sha256": "3072ad0b725ea3763ad4d1a7c744465f997614892dd1460f9cd815094ec61b70",
      "status": "PASS",
      "http": 200,
      "actual_bytes": 1037439,
      "actual_sha256": "3072ad0b725ea3763ad4d1a7c744465f997614892dd1460f9cd815094ec61b70",
      "snapshot_path": "/tmp/lab2-i01/snapshot/observations_b.jsonl"
    },
    {
      "source_bucket": "research-raw",
      "source_key": "lab2/inputs/batch-01/qa_reference.csv",
      "expected_bytes": 1423,
      "expected_sha256": "129463c2a4e63f2dabfa2a5dd31b92e4fe4858d2415d3c7ef495cc6c6d7b65c2",
      "status": "PASS",
      "http": 200,
      "actual_bytes": 1423,
      "actual_sha256": "129463c2a4e63f2dabfa2a5dd31b92e4fe4858d2415d3c7ef495cc6c6d7b65c2",
      "snapshot_path": "/tmp/lab2-i01/snapshot/qa_reference.csv"
    },
    {
      "source_bucket": "research-raw",
      "source_key": "lab2/inputs/batch-01/sensors.csv",
      "expected_bytes": 256,
      "expected_sha256": "02c390e127c37b9d12e9ca156eae23a450ad1ae037001cddd9651bb6516b9bde",
      "status": "PASS",
      "http": 200,
      "actual_bytes": 256,
      "actual_sha256": "02c390e127c37b9d12e9ca156eae23a450ad1ae037001cddd9651bb6516b9bde",
      "snapshot_path": "/tmp/lab2-i01/snapshot/sensors.csv"
    }
  ]
```

I02:

```
(.venv) PS C:\Users\Do Duc Kiet\Downloads\BigData_Lab02_Group4> Get-Content .\task1\evidence\I02.json
{
  "test": "I02",
  "mode": "LOCAL_PREFLIGHT",
  "source_counts": {
    "observations_a.csv": 5000,
    "observations_b.jsonl": 5205
  },
  "total_envelopes": 10205,
  "unique_physical_keys": 10205,
  "malformed_jsonl_preserved": 5,
  "status": "LOCAL-PASS"
}
```

I03

```
(.venv) PS C:\Users\Do Duc Kiet\Downloads\BigData_Lab02_Group4> Get-Content .\task1\evidence\integrity-test.json
{
  "test": "I03",
  "mode": "LOCAL_PREFLIGHT",
  "source_object": "observations_a.csv",
  "timestamp_utc": "2026-10-12T12:08:24.258483+00:00",
  "mutation": {
    "type": "single_byte_flip",
    "byte_offset": 0,
    "changed_bytes": 1
  },
  "expected_sha256": "5157418915d867980fac8d84551a8b856bba1669ca2e33cef91a15ee82101aea",
  "original_sha256": "5157418915d867980fac8d84551a8b856bba1669ca2e33cef91a15ee82101aea",
  "mutated_sha256": "abf16694706a4c8def6a7473a6fd14647309f5d8a4a52b70e2f4c59d245d4e7a",
  "integrity_error": "INPUT_INTEGRITY",
  "detected": true,
  "original_unchanged": true,
  "status": "LOCAL-PASS"
}
```

I04

```
(.venv) PS C:\Users\Do Duc Kiet\Downloads\BigData_Lab02_Group4> Get-Content .\task1\evidence\access-test.json
{
  "test_id": "I04",
  "principal": "ingestor",
  "timestamp_utc": "2026-10-11T17:32:53.289916+00:00",
  "tests": [
    {
      "name": "GET_RAW",
      "bucket": "research-raw",
      "key": "lab2/inputs/batch-01/sensors.csv",
      "http": 200,
      "bytes": 256,
      "status": "PASS"
    },
    {
      "name": "PUT_RELEASE",
      "bucket": "research-release",
      "key": "lab2/access-probes/i04-7d54aebd299d4670b8f6253e87c925b2.txt",
      "http": 403,
      "error": "AccessDenied",
      "status": "PASS"
    }
  ],
  "status": "PASS"
}
```

Task 2:

```
$ cd ~/lab01
$ ls snapshot quality_before.json trusted_manifest.json
$ python verify_task2.py trusted_manifest.json snapshot quality_before.json | t
ee verify_task2.txt
quality_before.json  trusted_manifest.json

snapshot:
observations_a.csv  observations_b.jsonl  qa_reference.csv  sensors.csv
PASS V1 integrity observations_a.csv
PASS V1 integrity observations_b.jsonl
PASS V1 integrity qa_reference.csv
PASS V1 integrity sensors.csv
PASS V2 inventory = 10205
PASS V2 parse failures  5
PASS V2 key rejected  40
PASS V2 key eligible  10160
PASS V2 winners W  9960
PASS V2 superseded  200
PASS V2 partition 5+40+200+W = raw
PASS V3 Q_REQUIRED  report 9850/9960 vs recomputed 9850/9960
PASS V4 Q_REQUIRED value & status
PASS V4 Q_REQUIRED has threshold/severity
PASS V3 Q_UNIQUE_KEY  report 9960/10160 vs recomputed 9960/10160
PASS V4 Q_UNIQUE_KEY value & status
PASS V4 Q_UNIQUE_KEY has threshold/severity
PASS V3 Q_VALUE  report 9710/9960 vs recomputed 9710/9960
PASS V4 Q_VALUE value & status
PASS V4 Q_VALUE has threshold/severity
PASS V3 Q_REGISTRY  report 9930/9960 vs recomputed 9930/9960
PASS V4 Q_REGISTRY value & status
PASS V4 Q_REGISTRY has threshold/severity
PASS V3 Q_CHRONO  report 9890/9960 vs recomputed 9890/9960
PASS V4 Q_CHRONO value & status
PASS V4 Q_CHRONO has threshold/severity
PASS V3 Q_TIMELY  report 9690/9890 vs recomputed 9690/9890
PASS V4 Q_TIMELY value & status
PASS V4 Q_TIMELY has threshold/severity
PASS V3 Q_QA_AGREEMENT  report 95/100 vs recomputed 95/100
PASS V4 Q_QA_AGREEMENT value & status
PASS V4 Q_QA_AGREEMENT has threshold/severity
PASS V3 Q_QA_COVERAGE  report 100/100 vs recomputed 100/100
PASS V4 Q_QA_COVERAGE value & status
PASS V4 Q_QA_COVERAGE has threshold/severity
PASS V5 no email in report
PASS V5 report has run_id and hashes

37/37 checks passed
```

```
DELL@DESKTOP-UHCRB1I MINGW64 ~/lab01
$ python -c "import json; d=json.load(open('quality_before.json')); print(d['populations']); [print(m['rule_id'], m['numerator'],'/',m['denominator'], m.get('observed_status')) for m in d['metrics']]"
{'raw': 10205, 'parse_failures': 5, 'key_rejected': 40, 'key_eligible': 10160, 'winners_W': 9960, 'superseded_duplicates': 200}
Q_REQUIRED 9850 / 9960 FAIL
Q_UNIQUE_KEY 9960 / 10160 FAIL
Q_VALUE 9710 / 9960 FAIL
Q_REGISTRY 9930 / 9960 FAIL
Q_CHRONO 9890 / 9960 FAIL
Q_TIMELY 9690 / 9890 PASS
Q_QA_AGREEMENT 95 / 100 PASS
Q_QA_COVERAGE 100 / 100 PASS
```

Task 3

```
=== TASK 3 INPUT ===
Total envelopes: 10205
Expected: 10205
Count check: PASS

=== SCHEMA ===
source_object : VARCHAR
source_row : BIGINT
parse_ok : BOOLEAN
canonical : STRUCT(record_id VARCHAR, sensor_id VARCHAR, event_time VARCHAR, ingest_time VARCHAR, reading VARCHAR, unit
VARCHAR, operator_email VARCHAR)
parse_error : VARCHAR
raw_line : VARCHAR

=== SAMPLE ENVELOPE ===
Top-level fields: ['source_object', 'source_row', 'parse_ok', 'canonical']
Canonical fields: ['record_id', 'sensor_id', 'event_time', 'ingest_time', 'reading', 'unit', 'operator_email']
```

```
=== TASK 3: PRECHECK ===
ELIGIBLE              10160
INGEST_PARSE             20
KEY                      20
PARSE                     5

=== RECONCILIATION ===
Total classified: 10205
Expected: 10205
PASS: All input envelopes accounted for
```

```
=== TASK 3: DEDUPLICATION ===
Raw envelopes:         10205
Precheck quarantine:   45
Winners:               9960
Duplicates:            200

=== RECONCILIATION ===
45 + 9960 + 200 = 10205
PASS: Conservation and unique winner IDs
```

```
=== contract.json ===
NOT FOUND

=== sensors.csv ===
Path: input\sensors.csv
Columns: ['sensor_id', 'site']
Path: task1\snapshot\sensors.csv
Columns: ['sensor_id', 'site']

=== qa_reference.csv ===
Path: input\qa_reference.csv
Columns: ['record_id', 'reference_c']
Path: task1\snapshot\qa_reference.csv
Columns: ['record_id', 'reference_c']
```

```
=== WINNER UNIT DISTRIBUTION ===
F                4960
C                4960
K                  40
```

```
=== TASK 3 CURATION ===
raw_envelopes        : 10205
curated              : 9610
quarantine           : 395
duplicates           : 200

=== WINNER VALIDATION REASONS ===
NUMERIC              : 60
RANGE                : 40
REFERENCE            : 30
REQUIRED             : 110
TIME_ORDER           : 30
TIME_PARSE           : 40
UNIT                 : 40
VALID                : 9610
```

```
    Directory: C:\Users\Do Duc Kiet\Downloads\BigData_Lab02_Group4\task3\candidate


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        10/12/2026     15:36         231597 curated.parquet
-a----        10/12/2026     15:36          13782 duplicates.csv
-a----        10/12/2026     15:36          19774 quarantine.csv
```


```
{
  "counts": {
    "raw_envelopes": 10205,
    "curated": 9610,
    "quarantine": 395,
    "duplicates": 200
  },
  "reconciled": true,
  "input_sha256": "e004df664e4cc7c24f0c5f45724f0202ebda987d2c433f408592152cdd81d5e1",
  "contract_sha256": "b25a48855fd318640ee74543eab9a1910be7a0f7c810b0db9b763d180decdf0d",
  "script_sha256": "5f1dbde6bc37f48b62bf680bab4b96b2d29fc9f42d3633744635513c55f3e6fe",
  "source_file_sha256": {
    "observations_a.csv": "5157418915d867980fac8d84551a8b856bba1669ca2e33cef91a15ee82101aea",
    "observations_b.jsonl": "3072ad0b725ea3763ad4d1a7c744465f997614892dd1460f9cd815094ec61b70"
  }
```


Test C01

```
=== C01: CONSERVATION ===
raw_count                : 10205
output_count             : 10205
missing_ids              : 0
invented_ids             : 0
repeated_output_ids      : 0
repeated_raw_ids         : 0
repeated_business_ids    : 0

C01: PASS
```

Test C02

```
=== C02: DEDUP REPLAY ===
winners                  : MATCH
duplicates               : MATCH
precheck_quarantine      : MATCH

C02 DEDUP: PASS
```

Test C03

```
=== C03: SCHEMA AND PRIVACY ===
PASS  record_id              VARCHAR
PASS  sensor_id              VARCHAR
PASS  site                   VARCHAR
PASS  event_time_utc         TIMESTAMP
PASS  ingest_time_utc        TIMESTAMP
PASS  temperature_c          DECIMAL(8,2)
PASS  is_late                BOOLEAN
PASS  source_object          VARCHAR
PASS  source_row             BIGINT
PASS  source_sha256          VARCHAR

Column count: 10
Privacy check: PASS
Invalid source hashes: 0

C03: PASS
```

Test C04

```
=== C04: EDGE CASE LOGIC CHECK ===
C04-01_fahrenheit                    PASS
C04-02_invalid_unit_numeric          PASS
C04-03_blank_email                   PASS
C04-04_invalid_newer_winner          PASS

Logic checks: 4/4
```

Test C05

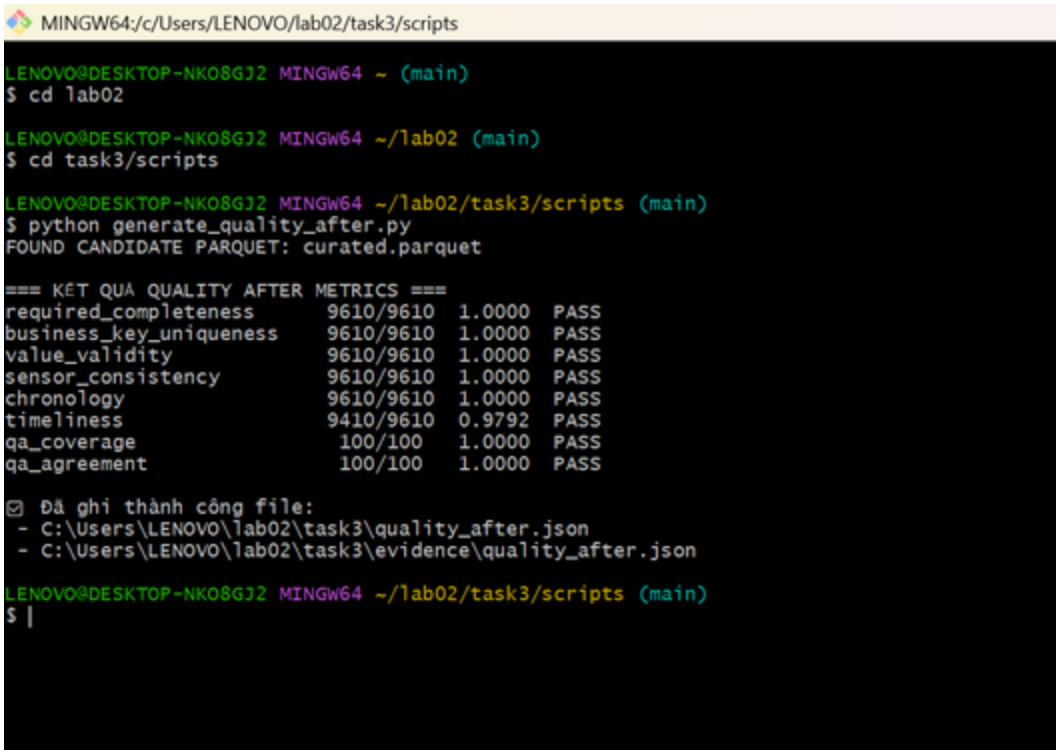
```
=== C05: CURATED OUTPUT CHECK ===
File: curated.parquet
Rows: 9610

=== SCHEMA ===
PASS record_id: VARCHAR (expected VARCHAR)
PASS sensor_id: VARCHAR (expected VARCHAR)
PASS site: VARCHAR (expected VARCHAR)
PASS event_time_utc: TIMESTAMP (expected TIMESTAMP)
PASS ingest_time_utc: TIMESTAMP (expected TIMESTAMP)
PASS temperature_c: DECIMAL(8,2) (expected DECIMAL(8,2))
PASS is_late: BOOLEAN (expected BOOLEAN)
PASS source_object: VARCHAR (expected VARCHAR)
PASS source_row: BIGINT (expected BIGINT)
PASS source_sha256: VARCHAR (expected VARCHAR)

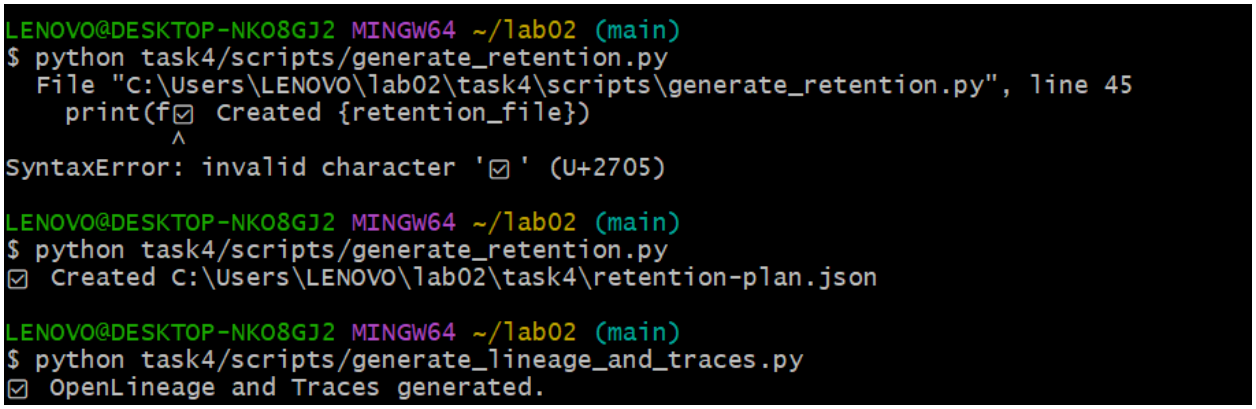
=== CHECKS ===
PASS correct_file_found
PASS nonempty_dataset
PASS exact_schema

Evidence: C:\Users\ADMIN\Downloads\BigData_Lab02_Group4\task3\evidence\C05.json
C05: PASS
```

Quality_after.json



Task 4



Thêm file readme

```
LENOVO@DESKTOP-NKO8GJ2 MINGW64 ~/lab02 (main)
$ cat << 'EOF' > task4/README.md
# Operational Metadata and Lineage Catalog

## Purpose & Synthetic Origin
This catalog and metadata suite operationalize data governance for the research telemetry pipeline. All datasets are derived from synthetic sensor measurements generated fo
r educational and benchmarking purposes under strict contract constraints.

## Files and Keys
- **curated.parquet**: The primary candidate dataset (`record_id` as primary key).
- **quarantine.parquet**: Isolated non-conforming records with primary failure reasons.
- **duplicates.csv**: Supporting duplicate ledger tracking deduplication actions.
- **catalog.json**: Machine-readable dataset catalog adhering to course profile requirements.

## Timestamps and Units
All timestamps (`event_time_utc`, `ingest_time_utc`, `as_of`) are represented in standard ISO 8601 UTC format. Temperature values are converted to standard Celsius (``C`) s
tored as `DECIMAL(8,2)`.

## Curation & Deduplication Policy
Incoming telemetry records are validated against contract rules. Business keys (`record_id`) undergo case-insensitive whitespace normalization (`UPPER(TRIM())`). Exact dupl
icates are deduplicated via deterministic ledger ordering, sending excess copies to the duplicate ledger while keeping valid unique entries.

## Quality Cohorts & Residual Limitations
Candidate quality is evaluated across eight distinct metrics (`Q_REQUIRED`, `Q_UNIQUE_KEY`, `Q_VALUE`, `Q_REGISTRY`, `Q_CHRONO`, `Q_TIMELY`, `Q_QA_COVERAGE`, `Q_QA_AGREEMEN
T`). Residual limitations include an allowable timeliness lag threshold of 95% (up to 900 seconds lag).

## Permitted Use & Access Boundary
This data is restricted strictly to synthetic teaching use. Production deployment or external re-distribution is prohibited. Raw and quarantine layers are classified as `RE
STRICTED`, whereas the curated layer is classified as `INTERNAL`.

## Software Versions & Reproduction Instructions
- **DuckDB**: >= 0.9.0
- **Python**: >= 3.10
Run execution scripts via:
`python task4/scripts/generate_catalog_and_queries.py`
`python task4/scripts/generate_retention.py`
`python task4/scripts/generate_lineage_and_traces.py`
All machine-readable evidence outputs are saved directly under `task4/evidence/` and `task4/lineage/`.
EOF
```

Task 5

```
LENOVO@DESKTOP-NKO8GJ2 MINGW64 ~/lab02 (main)
$ python task5/scripts/run_mutation_probes.py
☑ Executed 8 Mutation Probes (P01-P08) using trusted_manifest.

LENOVO@DESKTOP-NKO8GJ2 MINGW64 ~/lab02 (main)
$ python task5/scripts/publish_and_verify
C:\Users\LENOVO\AppData\Local\Programs\Python\Python312\python.exe: can't open file 'C:\\Users\\LENOVO\\lab02\\task5\\scripts\\publish_and_verify': [Errno 2] No such file o
r directory

LENOVO@DESKTOP-NKO8GJ2 MINGW64 ~/lab02 (main)
$ python task5/scripts/publish_and_verify.py
☑ Task 5 Release publication and Consumer verification completed successfully.

LENOVO@DESKTOP-NKO8GJ2 MINGW64 ~/lab02 (main)
$
```